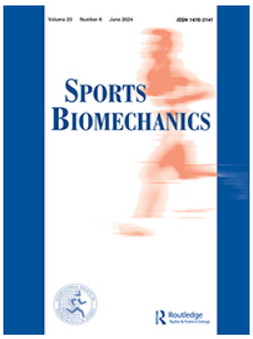

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Biceps femoris muscle-tendon strain during an entire overground sprint acceleration: a biomechanical explanation for hamstring injuries in the acceleration phase

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ABSTRACT

The objectives of this study were to analyse the peak muscle-tendon (MT) strain of the hamstring during an entire acceleration sprint overground and examine their relationship with relative joint angles and segment orientation in the sagittal plane, which are the direct causes of MT strain. Kinematic data were recorded using a 3D inertial motion capture system in 21 male semi-professional soccer players during 40-metre overground sprint. Scaled musculoskeletal models were used to estimate peak MT strain in the hamstring over 16 steps. Biceps femoris long head (BF^{LH}) exhibited the largest peaks in MT strain compared to semitendinosus (ST) and semimembranosus (SM) muscles across all the steps, with its overall strain decreased as the number of steps and maximum speed increased. Hip flexion angle was found to be a strong predictor ($p < 0.001$) of joint angles, being the orientation of the pelvis in the sagittal plane of the segment with the greatest influence ($p < 0.001$) on the peak MT strain of BF^{LH} during sprinting. The current study provides a biomechanical explanation for the high proportion of hamstring injuries in the acceleration phase of sprinting.

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Running biomechanics; musculoskeletal modelling; motion analysis; muscle injury; hamstring injury

Introduction

Performing soccer-specific actions at high speed is a crucial physical requirement for top-level soccer players. Short accelerations and linear sprints are two of the most significant actions in soccer, often preceding goals and other decisive moments (Faude et al., 2012). However, the majority of hamstring injuries (57–62%) occur during sprinting actions (Kenneally-Dabrowski et al., 2019; Woods et al., 2004). Hamstring muscle injuries are the most prevalent injuries in soccer, with their incidence increasing from 12% to 24% in the past two decades in men's professional soccer (Ekstrand et al., 2022). Thus, it is reasonable to expect that sprinting plays a pivotal role in soccer, in terms of both performance and injury risk.

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Previous studies in basic sciences (Garrett, 1990; Lieber & Friden, 1993) have been conducted to establish relationships between muscle-tendon (MT) mechanics and injury risk, with strain identified as a critical factor in muscle injury. In recent years, advanced musculoskeletal models and simulation techniques have facilitated the study and understanding of the behaviour of the hamstring MT complex during high-speed running and therefore provided the main understanding of injury mechanisms. The specific phase of sprinting in which hamstring injuries occur, whether it's the late swing or stance phase (Chumanov et al., 2012; Kenneally-Dabrowski et al., 2019), where the hamstring muscles are under tension while lengthening, remains a topic of debate. However, most modelling studies (Chumanov et al., 2011; Schache et al., 2012) and injury case studies (Heiderscheit et al., 2005; Schache et al., 2009, 2010) suggest that the late swing phase may be more vulnerable because of the increased mechanical strain on the hamstring MT complex.

During the late swing phase, the biceps femoris long head (BF^{LH}) exhibits a greater MT strain (10%) compared to the semitendinosus (8%) and semimembranosus (7%) muscles relative to their resting length (Chumanov et al., 2011; Schache et al., 2012) occurring with the maximum stretch of the contralateral iliacus and the second peak of anterior pelvic tilt (Chumanov et al., 2007; Franz et al., 2009; Nagano et al., 2014). This difference in the MT strain may contribute to the higher prevalence of BF^{LH} injuries compared to the semitendinosus (ST) and semimembranosus (SM) muscles (Ekstrand et al., 2022; Kenneally-Dabrowski et al., 2019; Woods et al., 2004). All this knowledge has become essential since the different practical proposals oriented to hamstring injury prevention (eccentric strength, flexibility, lumbopelvic control, neural tension, sprint technique) (Lahti et al., 2020) have had as their main common goal the control or reduction of BF^{LH} strain in sprinting. Computer simulations utilising musculoskeletal models have primarily focused on instrumented treadmill sprinting (Chumanov et al., 2011; Kenneally-Dabrowski et al., 2019; Schache et al., 2012), limiting their analysis to a few steps at either maximum or steady step speed. This approach has traditionally been employed to comprehend the mechanical role of the hamstrings during high-speed running. Nevertheless, the mechanical aspects of sprint acceleration performance differ from those of maximum-speed running (Higashihara et al., 2018), and although achieving maximal velocity is important in many team sports, the ability to accelerate and reach the highest velocity possible in the shortest period to gain the position or pass a defending player can be of far greater assistance to an athlete's on-field performance (Faude et al., 2012).

However, it overlooks the preceding acceleration phase, which has recently been highlighted as the primary sprint-related pattern in a retrospective analysis (Gronwald et al., 2022). Despite the strain factor has been considered as the main determinant of tissue failure by basic science studies the MT strain of the hamstring has not yet been analysed in detail during the acceleration phase (Garrett, 1990). A biomechanical explanation for the high proportion of hamstring injuries in the acceleration phase of sprinting is essential for developing both rehabilitation and prevention strategies based on this specific injury pattern.

If one of the primary goals of any preventive process is to focus on managing or reducing strain, it is crucial to identify its source. Furthermore, considering that the strain suffered by the hamstring muscles is influenced by the segments that this muscle

crosses, it is essential to describe the orientations of these (pelvis, femur and tibia) and their relationship with the MT strain of the hamstrings.

To achieve this, the objectives of this study were to (i) analyse the peaks in MT strain of the hamstring occurring during an entire sprint acceleration overground; (ii) establish the effect of hip flexion and knee flexion; and (iii) examine the orientation of the segments involved in generating strain in the BF^{LH} during sprinting, as they directly influence its strain. We hypothesised that BF^{LH} peak MT strain would be greater during the early steps of the sprint induced by the specific acceleration mechanics associated with the hip flexion angle and high anterior pelvic tilt, leading to an increased elongation of the hamstring muscle-tendon complex. These descriptions will provide valuable insights to enhance rehabilitation and prevention programmes specially for hamstring strain injuries occurring during the acceleration phase.

Materials and methods

Subjects

Twenty-one male semi-professional soccer players (mean ± SD, age, 22.90 ± 2.91 years; height, 1.82 ± 0.07 m; body mass, 75.29 ± 7.08 kg) were recruited for this study, which was approved by the University Ethics Committee of Rey Juan Carlos University (Approval Number: 0607202217622), and all participants gave their written informed consent before testing. All players were available for their teams and did not suffer any musculoskeletal injuries that could affect their sprinting performance at the time of testing. Subjects were not specifically excluded if they had a history of an acute hamstring muscle strain-type injury.

Experimental data collection

Data collection was conducted on a soccer field with natural grass. Before the testing session, a standardised specific warm-up was done under supervision. Kinematic data were recorded using a 3D inertial motion capture analysis system (MVN Link Xsens Technologies B.V., Enschede, the Netherlands) sampling at 240 Hz, using 17 inertial measurement units (IMUs) placed into a compression Lycra suit, which has been shown to provide valid angle estimates in lower/body sagittal kinematics in sports-specific situations in the field (Blair et al., 2018; Di Paolo et al., 2021; Nijmeijer et al., 2023). Each incased IMU contained a 3D gyroscope, a 3D accelerometer and a 3D magnetometer encased in an 18 g box (about the size of half a matchbox: 3.5 × 2.5 × 0.8 cm). A static pose calibration in N-pose position was completed for each subject prior to testing. All participants were instructed to perform two maximal 40-m sprints, starting from a static position with their preferred leg positioned slightly ahead in a 3-point start position. A 2-minute rest period was provided between trials.

Data analysis

All trials were reprocessed using the motion analysis software (MVN Analyze 2021.2, Xsens Technologies B.V., Enschede, the Netherlands) with high-definition reprocessing

in a no-level scenario, as recommend by the manufacturer (Schepers et al., 2018). This processing included filtering, skin artefact removal and joint estimation based on the MVN fusion engine (Schepers et al., 2018). The fastest trial for each athlete was selected for analysis, based on the sprint where the centre of mass (CoM) velocity reached its highest value. Segment orientations, joint angles and CoM kinematics were exported through MVN Analyze and imported into Python (v.3.10.2) to generate files for subsequent use in the muscle modelling software (v.4.2 OpenSim; Stanford University, USA) (Delp et al., 2007). Static pose calibration data were extracted from the motion tracking data and used to scale a generic model (Lai et al., 2017). This model consisted of 37 degrees of freedom and was driven by 80 Hill-type MT units across the pelvis and the lower limbs. In OpenSim, an inverse kinematics solver was used to produce a musculoskeletal simulation based on IMU data (DiRaimondo et al., 2022) using a weighted least squares method to determine the model's generalised orientations that best reproduced the IMU orientations. MT strain, defined as the change in MT length from an upright posture with all joint angles set to 0 in the specific model used, were expressed as a percentage of change. Kinematics were computed using the IK and then exported from OpenSim. Peaks in MT strain of BF^{LH}, SM and ST were identified for each step using Python (v.3.10.2), and hip flexion, knee flexion, and segment orientations (pelvis, femur and tibia) relative to the sagittal plane were computed at each peak MT strain of the BF^{LH}. All these variables were calculated up to the 16th step, where all the players reached between 95% and 99% of their maximum speed during the sprint (Figure 1). Gait cycles were obtained based on the Xsens MVN Analyze biomechanical model, which provided contact detection between foot and ground. Frames in which the foot and/or toes were in contact with the ground were identified from the foot contact data, and a segmentation of all the cycles in each trial was performed. Each individual

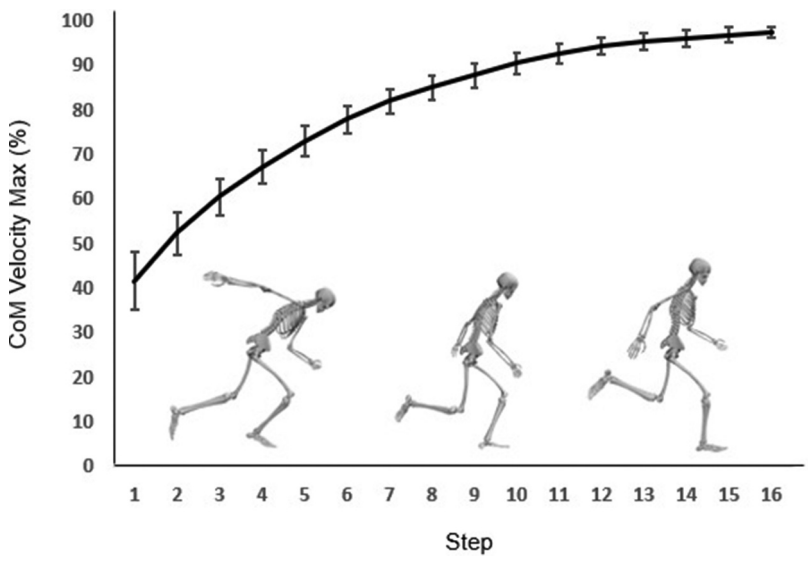

Figure 1. Mean and standard deviation of centre of mass (CoM) velocities are shown as a percentage relative to the maximum values attained at the peak muscle-tendon (MT) strain of biceps femoris long head (BF^{LH}) at each step during sprinting.

stride, for both the left and right side, was normalised to range from 0% to 100%. Once the cycles were normalised, the points at which the peak MT strain of the hamstring reached their maximum were marked as the gait % cycle. For cycle normalisation, the initial contact of the foot segment was considered the beginning (0%), and the next contact of the same foot was considered the end of the cycle (100%).

Statistical analysis

All the statistical analysis approaches described were performed using Jamovi v2.3 (Sydney, Australia).

To identify significant differences between the mean values of peak MT strain of the hamstring (BF^{LH}, SM, and ST) during sprinting, a repeated measures ANOVA followed by a Tukey post-hoc test was performed. The significance level was set at 5%. The assumption of normality of residual distribution was checked using the Shapiro-Wilk test.

Due to the structure of the dataset, three separate linear mixed-effects models (LMEMs) were conducted to examine the relationship between the peaks in MT strain of BF^{LH} and respective steps, joint angles (hip and knee flexion), and segment orientations in the sagittal plane (pelvis and femur) across all subjects at each step during sprinting. Each model treated subjects as clusters in which random effects varied. For the first LMEM, steps were used as the independent variable (fixed effect). For the second LMEM, joint angles (hip and knee flexion) were chosen as independent variables with fixed effects. Lastly, for the last LMEM, the segments forming part of the joint angle that exhibited the strongest correlation with peaks MT strain of BF^{LH} were selected as independent variables with fixed effects. The 95% confidence intervals of the fixed effects are reported for each model. The assumption of normality of residual distribution was checked using the Kolmogorov-Smirnov test for each model. The significance level was set at 5%.

A simple linear regression model was also performed using the mean values across subjects of anterior pelvic tilt with the mean peaks MT strain of BF^{LH} at each step during sprinting. The significance level was set at 5%.

Results

The average top sprinting speed for the cohort was 8.53 ± 0.36 m/s measured as CoM velocity. The mean times of peak MT strain for BF^{LH}, ST and SM during the gait cycle were 91.34% ± 1.70, 93.95% ± 1.77, 93.91% ± 1.67 respectively, across all overground sprints (Table 1a).

Peak MT strain of hamstrings across all the steps during sprinting

The analysis of peak MT hamstring strain revealed that the BF^{LH} experienced the greatest elongation across all the steps of the sprint, followed by the SM and ST (Figure 2).

Statistically significant differences were observed among the muscles ($p < 0.001$). Post-hoc analysis showed significant differences between BF^{LH} and SM, as well as between BF^{LH} and ST. No significant difference was found between SM and ST (Table 1b).

Table 1. a) Mean and standard deviation of peak muscle-tendon (MT) hamstring strain and their timing during sprinting. Peak MT hamstring lengths are calculated relative to their lengths in an upright configuration. b) post-hoc Tukey test results between all peak MT hamstring lengths.

Muscle-Tendon		Peak MT strain (%)			Timing of peak (%gait cycle)	
a)						
BF^{LH}		9.03 ± 1.42			91.34 ± 1.70	
ST		6.09 ± 1.62			93.95 ± 1.77	
SM		6.10 ± 1.43			93.91 ± 1.67	
Comparison between Peak MT strain		Mean Difference	SE	df	t	pTukey
b)						
BF^{LH}	ST	2.94	0.08	20.0	35.23	**<.001**
	SM	2.93	0.05	20.0	55.75	**<.001**
ST	SM	−0.01	0.05	20.0	−0.22	0.974

Significant results are highlighted in bold. BFLH: Biceps femoris long head. ST: Semitendinosus SM: Semimembranosus. SE: Standard Error. df: Degrees of freedom. t: t-value.

Linear mixed effect model output for steps and peak MT strain of BF^{LH} during sprinting

In the first LMEM (Table 2a), the steps were found to be a significant predictor ($p < 0.001$) for the peak MT strain of BF^{LH} during sprinting, with a fixed effects estimate of – 0.05 (Table 2a). The model had an Akaike information criteria (AIC) of 816.9.

Linear mixed effect model output for hip and knee flexion joint angles for peak MT strain of BF^{LH} during sprinting

In the second LMEM (Table 2b), hip and knee flexion were significant predictors ($p < 0.001$) of the peak MT strain of BF^{LH} during sprinting, with a fixed effects estimate for the hip flexion of 0.21 and – 0.11 for the knee flexion. The model had an AIC of 168.2.

Linear mixed effect model output for pelvis and femur orientation in the sagittal plane and peak MT strain of BF^{LH} during sprinting

The third linear mixed-effects model (Table 2c) showed that only pelvis segment orientation in the sagittal plane (anterior pelvic tilt) was a significant predictor ($p < 0.001$) for the peak MT strain of BF^{LH} during sprinting, with a fixed effects estimate of – 0.05. The model had an AIC of 766.2.

Simple linear model for anterior pelvic tilt and peak MT strain of BF^{LH} during sprinting

A significant interaction ($p < 0.001$) with a negative strong correlation (r: −0.86) was found between anterior pelvic tilt and BF^{LH} peak MT strain during sprinting (Figure 3).

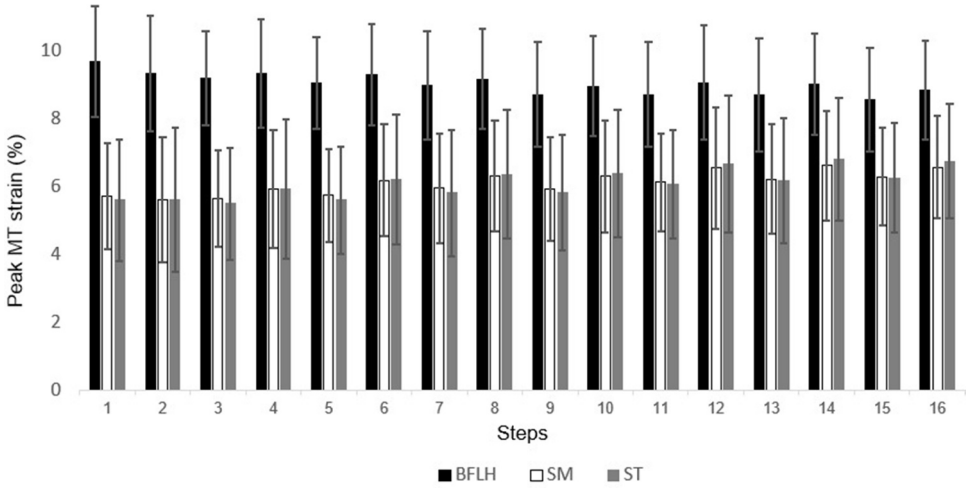

Figure 2. Mean peak muscle-tendon (MT) strain and standard deviation of the biceps femoris long head (BF^{LH}), semitendinosus (ST), and semimembranosus (SM) are shown across all steps during sprinting. Peak MT hamstring strain are calculated relative to their lengths in an upright configuration.

Discussion and implication

The use of advanced musculoskeletal models and simulation techniques has provided fundamental understanding of the behaviour of the hamstring MT complex during high-speed running and has offered preliminary insights into potential injury mechanistic underpinnings. The aim of the present study was to describe and quantify the peak MT strain of the hamstring muscles, as well as the causes, across all the steps during an entire sprint acceleration in soccer players. The main findings of the present study were: (i) BF^{LH} showed the greatest MT strain among the muscles

Table 2. a) linear mixed effect model information for MT strain peaks of BF^{LH} for steps during sprinting. equation: BF^{LH} MT strain = 9.03–0.05 × step b) linear mixed effect model information for peak MT strain of BF^{LH} for hip and knee joint flexion during sprinting. Equation: BF^{LH} MT strain = 9.03 + 0.21 × hip flexion – 0.11 × knee flexion. c) linear mixed effect model information for peak MT strain of BF^{LH} for femur and pelvis segment orientation in the sagittal plane during sprinting. Equation: BF^{LH} MT strain = 9.03–0.05 × pelvis orientation – 0.01 × femur orientation.

Predictor	Estimate	SE	CI		df	t	p
a)							
Intercept	9.03	0.31	8.42	9.64	20.0	29.06	**<.001**
Step	−0.05	0.01	−0.07	−0.03	314.0	−5.91	**<.001**
b)							
Intercept	9.03	0.05	8.94	9.12	18.4	194.0	**<.001**
Hip flexion	0.21	0.0	0.20	0.22	162.6	51.8	**<.001**
Knee flexion	−0.11	0.0	−0.12	−0.11	188.1	−45.2	**<.001**
c)							
Intercept	9.03	0.27	8.50	9.55	19.8	33.78	**<.001**
Pelvis	−0.05	0.01	−0.06	−0.04	319.2	−9.79	**<.001**
Femur	−0.01	0.01	−0.03	0.01	325.2	−0.76	0.45

Significant results are highlighted in bold. SE: Standard Error. CI: Confidence Interval. df: Degrees of Freedom. t: t-value.

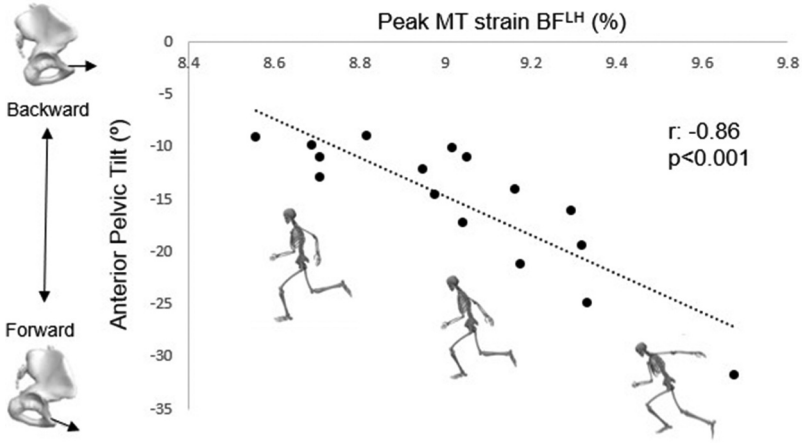

Figure 3. Simple linear correlation between anterior pelvic tilt and peak muscle-tendon (MT) strain of the biceps femoris long head (BF^{LH}) is shown across all 16 steps.

that constitute the hamstring musculature during the entire sprint acceleration. ii) The peak MT strain of BF^{LH} decreased as speed and step number increased, suggesting a greater strain in the muscle fibres during early acceleration compared to the maximum velocity phases. iii) The hip flexion angle had a greater influence on the resulting peak MT strain of BF^{LH} than knee flexion, with the pelvis orientation in the sagittal plane being the segment that had the most significant influence on the peak MT strain during sprinting.

During the entire sprint acceleration, the peak MT strain of the hamstring occurred during the late swing phase (around 92% of the gait cycle), with the largest peak observed in the MT strain of the BF^{LH} compared to ST and SM. These results were similar to those reported in previous studies that analysed only a few steps close to top speed on an instrumented treadmill (Chumanov et al., 2011; Kenneally-Dabrowski et al., 2019; Schache et al., 2012). Experimental studies and musculoskeletal models applied to sprinting have explained the greater MT length found in BF^{LH} compared to the medial hamstring by the combination of a longer hip moment arm and a smaller knee moment arm (Thelen et al., 2005; Visser et al., 1990). Specifically, as the hip angle remains high throughout the sprint and exerts a significant impact on the length of BF^{LH}, while there is a smaller reduction in the overall length of this muscle due to the small knee moment arm (Thelen et al., 2005; Visser et al., 1990), the net result of these combined effects is an increased strain demand throughout the BF^{LH} during the entire sprint acceleration compared to ST and SM. These findings, along with those provided by in vitro (Garrett, 1990; Lieber & Friden, 1993) and in vivo (Guilhem et al., 2016) studies, suggest that the magnitude of stretching applied to the muscle fascicles of the biceps femoris, combined with the high torque produced at extreme lengths during the sprint, may explain the greater muscle damage. This is also related to the higher prevalence of lateral hamstring strain injuries compared to medial ones. The recording of kinematics for each step throughout the sprint in this study enabled a detailed analysis of the BF^{LH} strain behaviour during all sprint phases. The results revealed a significant reduction in the

peak MT strain of BF^{LH} as speed increased, indicating greater strain during early acceleration compared to high-speed running phases (Figure 4).

These findings can be associated with the differences in kinematics and kinetic demands between the two phases. During acceleration, a greater lean in trunk posture and a high hip flexion angle are observed (Nagahara et al., 2018), allowing the production of greater horizontal forces required for optimal acceleration (Morin et al., 2011, 2015). In contrast, high-speed running is characterised by a more upright posture, which is associated with the ability to generate high amounts of ground reaction force in the vertical direction (Clark & Weyand, 2014; Clark et al., 2017). Taking this into account and considering a previous in vivo study (Di Raimondo et al., 2022) showing the length of BF^{LH} to be more sensitive to changes in hip angle (due to a greater moment arm) compared to the knee, it is plausible to explain the greater strain observed during sprint acceleration compared to high-speed running. This could be attributed to the differing hip flexion and knee angles (Figure 5) between these two phases, highlighting the significance of variations in these angles during different sprinting phases. The significantly higher strain observed in the BF^{LH} during the initial steps combined with the high mechanical demand imposed on the posterior thigh muscles (Mendiguchia et al., 2014, 2016; Morin et al., 2015), may explain the findings of a recent retrospective study in soccer (Gronwald et al., 2022). This study observed that more than half of the sprint-related hamstring injuries (all involving biceps femoris) occurred during linear sprint acceleration rather than maximum velocity scenarios. Additionally, the increasing trend of shorter, faster, and more frequent sprints in professional soccer (Pons et al., 2021), along with their impact on soccer performance (Faude et al., 2012), may contribute to the

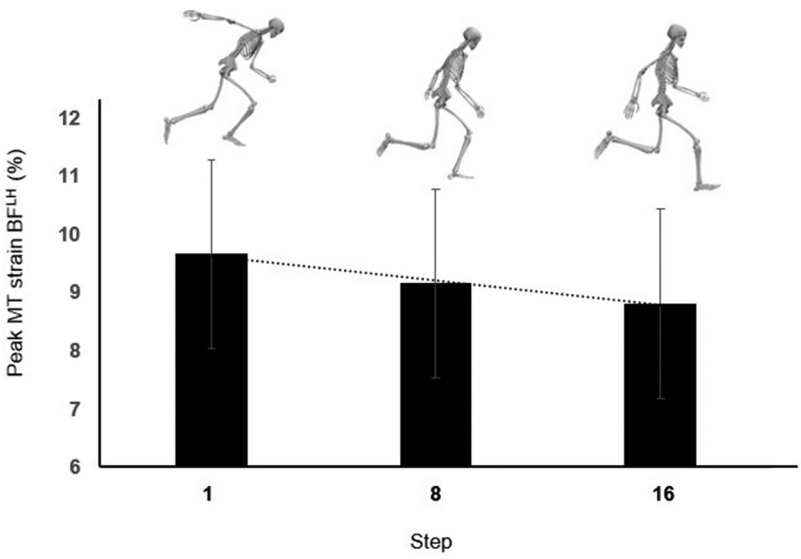

Figure 4. Example of the mean peak of the muscle-tendon (MT) strain of the biceps femoris long head (BF^{LH}) and standard deviation during the first, eighth and sixteenth steps during sprinting. Dotted lines represent the linear trend line.

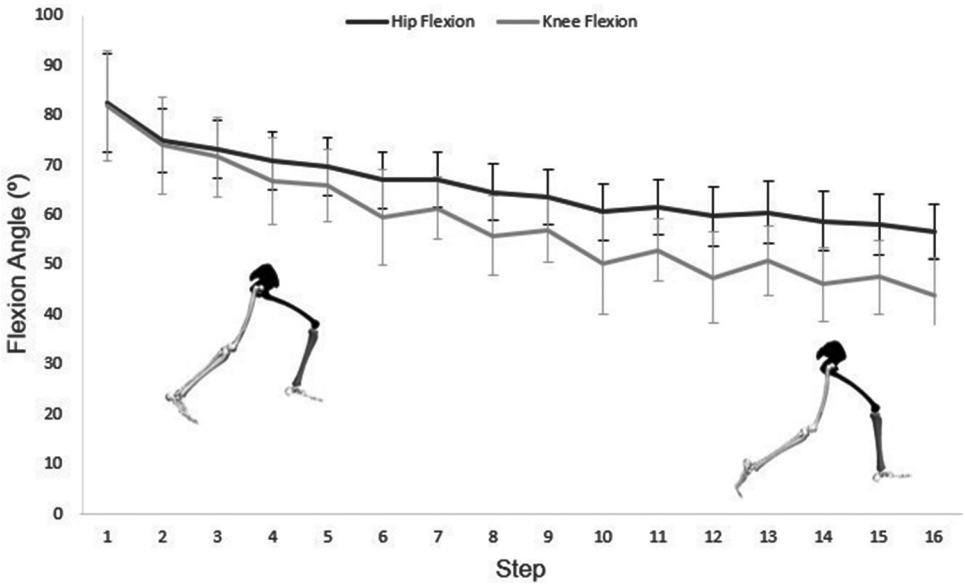

Figure 5. Mean and standard deviation of hip flexion and knee flexion in the peak muscle-tendon (MT) strain of BF^{LH} are shown across all steps during sprinting. The skeleton on the left visually represents the kinematics observed during the acceleration phase, while the skeleton on the right represents the kinematics observed during the maximum velocity phase.

exponential increase in the proportion of diagnosed hamstring injuries in the last 20 years (from 12% to 24%) (Ekstrand et al., 2022).

Finally, it remains to be determined whether the pelvis or femur segment contributes more significantly to the peak MT strain of the BF^{LH}, as the hip joint angle is defined by the angle formed between the pelvis and femur. The linear mixed model employed in this study showed that pelvis orientation in the sagittal plane was the most significant predictor component segment compared to femur orientation in determining the peak MT strain of the BF^{LH} during sprinting. This finding was recently confirmed by an in vivo study (Higashihara et al., 2015) that analysed the effect of sprinting with trunk forward lean and anterior pelvic tilt, and an in vitro cadaveric study (Mendiguchia et al., 2024) where the significant impact of anterior pelvic tilt on hamstring length was demonstrated when the femur remained fixed.

Current musculoskeletal models estimate the MT parameters based on anatomical measurements of cadaver specimens and magnetic resonance images (Rajagopal et al., 2016), being hamstring MT length estimations during sprinting based on hip and knee joint angles, giving the same weight to all the segments involved. However, this approach may underestimate the strain impact of the posterior and superior translation of the ischial tuberosity (the segment in which the hamstrings are directly inserted) during anterior pelvic tilt movements, as recently demonstrated (Mendiguchia et al., 2024). By giving equal weight to the movement of the pelvis and the femur in hamstring strain, musculoskeletal models may overlook the specific contribution of pelvic movement. These findings help explain the association between anterior pelvic tilt and hamstring injury risk

observed in both prospective and retrospective studies (Daly et al., 2016; Schuermans et al., 2017).

In summary, the findings of this study suggest that hamstring injuries occurring during the acceleration phase may be closely linked to the biceps femoris muscle-tendon strain, particularly caused by the high hip flexion angle induced by the high anterior pelvic tilt during the initial steps. Considering that exposure to explosive accelerations is more frequent (Pons et al., 2021) than maximum speed during soccer practice, the combination of high biceps femoris electromyography activation (Higashihara et al., 2018), increased mechanical demand on the hamstring muscles (Morin et al., 2011, 2015) to drive the body powerfully in the forward direction (Jacobs & van Ingen Schenau, 1992) and the aforementioned greater strain observed in the BF^{LH} during the early acceleration phase creates an ideal environment for the high incidence of biceps femoris injuries in soccer.

Translating the findings of this study into clinical practice would involve targeting the pelvis as a key area when the objective of rehabilitation and/or prevention is to reduce biceps femoris strain during sprinting. A recent multimodal intervention (Mendiguchia et al., 2022) combining lumbopelvic control exercises with a running technique programme has shown promising results in reducing anterior pelvic tilt (by 5º) over a 6-week period to control and mitigate strain during maximal sprinting. During the rehabilitation process, exercises or drills that involve leaning the trunk forward or achieving a high level of hip flexion, especially when induced by an anterior pelvic tilt, could be employed cautiously if the goal is to mitigate potential excessive strain on the injured musculature, particularly in hamstring injuries occurring during the acceleration phase, due to their potential connection with the strain factor.

Limitations

It is, however, important to recognise some limitations in this study according to the use of musculoskeletal models. MT strain also depends on dynamic interaction between muscle and tendon, and this relationship also important in understanding myotendinous and muscle injuries and the behaviour of this complex structure. During this study muscle activation and force were not measured. Therefore, analysing fibre/fascicle behaviour would be more appropriate than analysing MT strain, as studying its behaviour may not accurately reflect fibre strain. In addition to muscle kinematics analysis, it would be advantageous to investigate all differences in muscle forces during an entire sprint acceleration overground.

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References

Blair, S., Duthie, G., Robertson, S., Hopkins, W., & Ball, K. (2018). Concurrent validation of an inertial measurement system to quantify kicking biomechanics in four football codes. *Journal of Biomechanics*, *73*, 24–32. https://doi.org/10.1016/j.jbiomech.2018.03.031

Chumanov, E. S., Heiderscheit, B. C., & Thelen, D. G. (2007). The effect of speed and influence of individual muscles on hamstring mechanics during the swing phase of sprinting. *Journal of Biomechanics*, *40*(16), 3555–3562. https://doi.org/10.1016/j.jbiomech.2007.05.026

Chumanov, E. S., Heiderscheit, B. C., & Thelen, D. G. (2011). Hamstring musculotendon dynamics during stance and swing phases of high-speed running. *Medicine and Science in Sports and Exercise*, *43*(3), 525–532. https://doi.org/10.1249/MSS.0b013e3181f23fe8

Chumanov, E. S., Schache, A. G., Heiderscheit, B. C., & Thelen, D. G. (2012). Hamstrings are most susceptible to injury during the late swing phase of sprinting. *British Journal of Sports Medicine*, *46*(2), 90. https://doi.org/10.1136/bjsports-2011-090176

Clark, K. P., Ryan, L. J., & Weyand, P. G. (2017). A general relationship links gait mechanics and running ground reaction forces. *The Journal of Experimental Biology*, *220*(Pt 2), 247–258. https://doi.org/10.1242/jeb.138057

Clark, K. P., & Weyand, P. G. (2014). Are running speeds maximized with simple-spring stance mechanics? *Journal of Applied Physiology*, *117*(6), 604–615. https://doi.org/10.1152/japplphysiol.00174.2014

Daly, C., Persson, U. M., Twycross-Lewis, R., Woledge, R. C., & Morrissey, D. (2016). The biomechanics of running in athletes with previous hamstring injury: A case-control study. *Scandinavian Journal of Medicine & Science in Sports*, *26*(4), 413–420. https://doi.org/10.1111/sms.12464

Delp, S. L., Anderson, F. C., Arnold, A. S., Loan, P., Habib, A., John, C. T., Guendelman, E., & Thelen, D. G. (2007). OpenSim: Open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, *54*(11), 1940–1950. https://doi.org/10.1109/TBME.2007.901024

Di Paolo, S., Lopomo, N. F., Della Villa, F., Paolini, G., Figari, G., Bragonzoni, L., Grassi, A., & Zaffagnini, S. (2021). Rehabilitation and return to sport assessment after anterior cruciate ligament injury: Quantifying joint kinematics during complex high-speed tasks through wearable sensors. *Sensors (Basel)*, *21*(7), 2331. https://doi.org/10.3390/s21072331

Di Raimondo, G., Vanwanseele, B., van der Have, A., Emmerzaal, J., Willems, M., Killen, B. A., & Jonkers, I. (2022). Inertial sensor-to-segment calibration for accurate 3D joint angle calculation for use in OpenSim. *Sensors (Basel)*, *22*(9), 3259. https://doi.org/10.3390/s22093259

Ekstrand, J., Bengtsson, H., Walden, M., Davison, M., Khan, K. M., & Hagglund, M. (2022). Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: The UEFA Elite Club injury study from 2001/02 to 2021/22. *British Journal of Sports Medicine*, *57*(5), 292–298. https://doi.org/10.1136/bjsports-2021-105407

Faude, O., Koch, T., & Meyer, T. (2012). Straight sprinting is the most frequent action in goal situations in professional football. *Journal of Sports Sciences*, *30*(7), 625–631. https://doi.org/10.1080/02640414.2012.665940

Franz, J. R., Paylo, K. W., Dicharry, J., Riley, P. O., & Kerrigan, D. C. (2009). Changes in the coordination of hip and pelvis kinematics with mode of locomotion. *Gait & Posture*, *29*(3), 494–498. https://doi.org/10.1016/j.gaitpost.2008.11.011

Garrett, W. E., Jr. (1990). Muscle strain injuries: Clinical and basic aspects. *Medicine and Science in Sports and Exercise*, *22*(4), 436–443. https://doi.org/10.1249/00005768-199008000-00003

Gronwald, T., Klein, C., Hoenig, T., Pietzonka, M., Bloch, H., Édouard, P., & Hollander, K. (2022). Hamstring injury patterns in professional male football (soccer): A systematic video analysis of 52 cases. *British Journal of Sports Medicine*, *56*(3), 165–171. https://doi.org/10.1136/bjsports-2021-104769

Guilhem, G., Doguet, V., Hauraix, H., Lacourpaille, L., Jubeau, M., Nordez, A., & Dorel, S. (2016). Muscle force loss and soreness subsequent to maximal eccentric contractions depend on the amount of fascicle strain in vivo. *Acta Physiologica*, *217*(2), 152–163. https://doi.org/10.1111/apha.12654

Heiderscheit, B. C., Hoerth, D. M., Chumanov, E. S., Swanson, S. C., Thelen, B. J., & Thelen, D. G. (2005). Identifying the time of occurrence of a hamstring strain injury during treadmill running: A case study. *Clin Biomech (Bristol, Avon)*, *20*(10), 1072–1078. https://doi.org/10.1016/j.clinbiomech.2005.07.005

Higashihara, A., Nagano, Y., Ono, T., & Fukubayashi, T. (2018). Differences in hamstring activation characteristics between the acceleration and maximum-speed phases of sprinting. *Journal of Sports Sciences*, *36*(12), 1313–1318. https://doi.org/10.1080/02640414.2017.1375548

Higashihara, A., Nagano, Y., Takahashi, K., & Fukubayashi, T. (2015). Effects of forward trunk lean on hamstring muscle kinematics during sprinting. *Journal of Sports Sciences*, *33*(13), 1366–1375. https://doi.org/10.1080/02640414.2014.990483

Jacobs, R., & van Ingen Schenau, G. J. (1992). Intermuscular coordination in a sprint push-off. *Journal of Biomechanics*, *25*(9), 953–965. https://doi.org/10.1016/0021-9290(92)90031-U

Kenneally-Dabrowski, C. J. B., Brown, N. A. T., Lai, A. K. M., Perriman, D., Spratford, W., & Serpell, B. G. (2019). Late swing or early stance? A narrative review of hamstring injury mechanisms during high-speed running. *Scandinavian Journal of Medicine & Science in Sports*, *29*(8), 1083–1091. https://doi.org/10.1111/sms.13437

Lahti, J., Mendiguchia, J., Ahtiainen, J., Anula, L., Kononen, T., Kujala, M., Matinlauri, A., Peltonen, V., Thibault, M., Toivonen, R.-M., Edouard, P., & Morin, J. B. (2020). Multifactorial individualised programme for hamstring muscle injury risk reduction in professional football: Protocol for a prospective cohort study. *BMJ Open Sport and Exercise Medicine*. Published online 2020 *6*(1), e000758. https://doi.org/10.1136/bmjsem-2020-000758

Lai, A. K. M., Arnold, A. S., & Wakeling, J. M. (2017). Why are antagonist muscles co-activated in my simulation? a musculoskeletal model for analysing human locomotor tasks. *Annals of Biomedical Engineering*, *45*(12), 2762–2774. https://doi.org/10.1007/s10439-017-1920-7

Lieber, R. L., & Friden, J. (1993). Muscle damage is not a function of muscle force but active muscle strain. *Journal of Applied Physiology: Respiratory, Environmental and Exercise Physiology*, *74*(2), 520–526. https://doi.org/10.1152/jappl.1993.74.2.520

Mendiguchia, J., Aranzazu Garrues, M., Schilders, E., Myer, G. D., & Dalmau-Pastor, M. (2024). Anterior pelvic tilt increases hamstring strain and is a key factor to target for injury prevention and rehabilitation. *Knee Surgery, Sports Traumatology, Arthroscopy: Official Journal of the ESSKA*, *32*(3), 573–582. https://doi.org/10.1002/ksa.12045

Mendiguchia, J., Castano-Zambudio, A., Jimenez-Reyes, P., Morin, J., Edouard, P., Conceição, F., Tawiah-Dodoo, J., & Colyer, S. L. (2022). Can we modify maximal speed running posture? Implications for performance and hamstring injury management. *International Journal of Sports Physiology and Performance*, *17*(3), 374–383. https://doi.org/10.1123/ijspp.2021-0107

Mendiguchia, J., Edouard, P., Samozino, P., Brughelli, M., Cross, M., Ross, A., Gill, N., & Morin, J. B. (2016). Field monitoring of sprinting power–force–velocity profile before, during and after hamstring injury: Two case reports. *Journal of Sports Sciences*, *34*(6), 535–541. https://doi.org/10.1080/02640414.2015.1122207

Mendiguchia, J., Samozino, P., Martinez-Ruiz, E., Brughelli, M., Schmikli, S., Morin, J.-B., & Mendez-Villanueva, A. (2014). Progression of mechanical properties during on-field sprint running after returning to sports from a hamstring muscle injury in soccer players. *International Journal of Sports Medicine*, *35*(8), 690–695. https://doi.org/10.1055/s-0033-1363192

Morin, J. B., Edouard, P., & Samozino, P. (2011). Technical ability of force application as a determinant factor of sprint performance. *Medicine and Science in Sports and Exercise*, *43*(9), 1680–1688. https://doi.org/10.1249/MSS.0b013e318216ea37

Morin, J. B., Gimenez, P., Edouard, P., Arnal, P., Jiménez-Reyes, P., Samozino, P., Brughelli, M., & Mendiguchia, J. (2015). Sprint acceleration mechanics: The major role of hamstrings in horizontal force production. *Frontiers in Physiology*, *6*, 404. https://doi.org/10.3389/fphys.2015.00404

Nagahara, R., Matsubayashi, T., Matsuo, A., & Zushi, K. (2018). Kinematics of the thorax and pelvis during accelerated sprinting. *The Journal of Sports Medicine and Physical Fitness*, *58*(9), 1253–1263. https://doi.org/10.23736/S0022-4707.17.07137-7

Nagano, Y., Higashihara, A., Takahashi, K., & Fukubayashi, T. (2014). Mechanics of the muscles crossing the hip joint during sprint running. *Journal of Sports Sciences*, *32*(18), 1722–1728. https://doi.org/10.1080/02640414.2014.915423

Nijmeijer, E. M., Heuvelmans, P., Bolt, R., Gokeler, A., Otten, E., & Benjaminse, A. (2023). Concurrent validation of the xsens IMU system of lower-body kinematics in jump-landing and change-of-direction tasks. *Journal of Biomechanics*, *154*, 111637. https://doi.org/10.1016/j.jbiomech.2023.111637

Pons, E., Ponce-Bordon, J. C., Diaz-Garcia, J., López Del Campo, R., Resta, R., Peirau, X., & García-Calvo, T. (2021). A longitudinal exploration of match running performance during a football match in the Spanish La Liga: A four-season study. *International Journal of Environmental Research and Public Health*, *18*(3), 1133. https://doi.org/10.3390/ijerph18031133

Rajagopal, A., Dembia, C. L., MS, D., Delp, D. D., Hicks, J. L., & Delp, S. L. (2016). Full-body musculoskeletal model for muscle-driven simulation of human gait. *IEEE Transactions on Bio-Medical Engineering*, *63*(10), 2068–2079. https://doi.org/10.1109/TBME.2016.2586891

Schache, A. G., Dorn, T. W., Blanch, P. D., Brown, N. A., & Pandy, M. G. (2012). Mechanics of the human hamstring muscles during sprinting. *Medicine and Science in Sports and Exercise*, *44*(4), 647–658. https://doi.org/10.1249/MSS.0b013e318236a3d2

Schache, A. G., Kim, H. J., Morgan, D. L., & Pandy, M. G. (2010). Hamstring muscle forces prior to and immediately following an acute sprinting-related muscle strain injury. *Gait & Posture*, *32*(1), 136–140. https://doi.org/10.1016/j.gaitpost.2010.03.006

Schache, A. G., Wrigley, T. V., Baker, R., & Pandy, M. G. (2009). Biomechanical response to hamstring muscle strain injury. *Gait & Posture*, *29*(2), 332–338. https://doi.org/10.1016/j.gaitpost.2008.10.054

Schepers, M., Giuberti, M., & Bellusci, G. (2018). Xsens MVN: Consistent tracking of human motion using inertial sensing. *Xsens Technol*, 1–8. https://doi.org/10.13140/RG.2.2.22099.07205

Schuermans, J., Van Tiggelen, D., Palmans, T., Danneels, L., & Witvrouw, E. (2017). Deviating running kinematics and hamstring injury susceptibility in male soccer players: Cause or consequence? *Gait & Posture*, *57*, 270–277. https://doi.org/10.1016/j.gaitpost.2017.06.268

Thelen, D. G., Chumanov, E. S., Hoerth, D. M., Best, T. M., Swanson, S. C., Li, L. I., Young, M., & Heiderscheit, B. C. (2005). Hamstring muscle kinematics during treadmill sprinting. *Medicine and Science in Sports and Exercise*, *37*(1), 108–114. https://doi.org/10.1249/01.MSS.0000150078.79120.C8

Visser, J. J., Hoogkamer, J. E., Bobbert, M. F., & Huijing, P. A. (1990). Length and moment arm of human leg muscles as a function of knee and hip-joint angles. *European Journal of Applied Physiology and Occupational Physiology*, *61*(5–6), 453–460. https://doi.org/10.1007/BF00236067

Woods, C., Hawkins, R. D., Maltby, S., Hulse, M., Thomas, A., & Hodson, A. (2004). The Football Association Medical Research Programme: An audit of injuries in professional football–analysis of hamstring injuries. *British Journal of Sports Medicine*, *38*(1), 36–41. https://doi.org/10.1136/bjsm.2002.002352